



A Graduation Project entitled:

LAPSTORE

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1. Project Proposal

1.1. Overview

LapStore is an e-commerce platform offering a wide range of laptops at competitive prices, providing comprehensive product information, excellent user experience, and outstanding customer support.

1.2. Objectives

- Deliver a seamless user experience, increase sales, and improve customer satisfaction.

1.3. Scope

- Develop a website with an easy-to-use interface, product management system, and shopping cart functionality.

2. Project Plan

2.1. Timeline (Gantt Chart)

Phase	Task	Duration
Phase 1	Search for Bootstrap Template and Install	1 Day
Phase 2	Design ERD and Relationships	3 Days
Phase 3	Database Normalization	1 Day
Phase 4	Create Migration using Entity Framework	3 Days
Phase 5	Frontend Development	2 Weeks
Phase 6	Backend Development	6 Weeks

Phase 7	Testing & Debugging	1 Week
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2.2. Milestones & Deliverables

Completion of design, development, testing, and final delivery.

3. Task Assignment & Roles

Role	Responsibility
Backend Developer and Team Lead	Develop APIs, Data access Layer, and business logic Layer.
Frontend Developer	Implement UI components and user interactions.
Database Designer	Design ERD, and schema
Software Tester	Create unit tests, ensures quality, and validates software.

4. Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation Strategy
Delays in development	High	Regular progress tracking, buffer time in schedule
Security vulnerabilities	High	Implement secure coding practices, regular audits
Payment gateway failures	Medium	Use multiple payment providers
Poor user adoption	Medium	Marketing campaigns, user-friendly design

5. Key Performance Indicators (KPIs)

KPI	Description
System Uptime	Ensuring 99% uptime of the platform.
Response Time	API response time < 500ms
User Adoption Rate	Growth in new user registrations.
Order Completion Rate	Percentage of successful transactions.
Customer Satisfaction	Measured via reviews and feedback.

6. Requirements Gathering

6.1 Stakeholder Analysis:

Identifying key stakeholders and their needs:

- Buyers – Require an easy-to-use interface, secure payments, and reliable order tracking.
- Sellers – Need a store management system and marketing tool
- Administrators – Require full control over platform operations, user management.

6.2 User Stories & Use Cases

Buyer User Stories

- As a buyer, I want to search for laptops easily so that I can find what I need quickly.
- As a buyer, I want to securely complete my payment so that I feel safe making transactions.
- As a buyer, I want to track my orders so that I know when they will arrive.

Seller User Stories

- As a seller, I want to manage my products and inventory so that I can keep my store updated.
- As a seller, I want to analyze my sales performance so that I can optimize my business.

Admin User Stories

- As an admin, I want to manage user accounts so that I can maintain platform integrity.
- As an admin, I want to resolve disputes so that I can ensure smooth transactions.

6.3 Functional Requirements

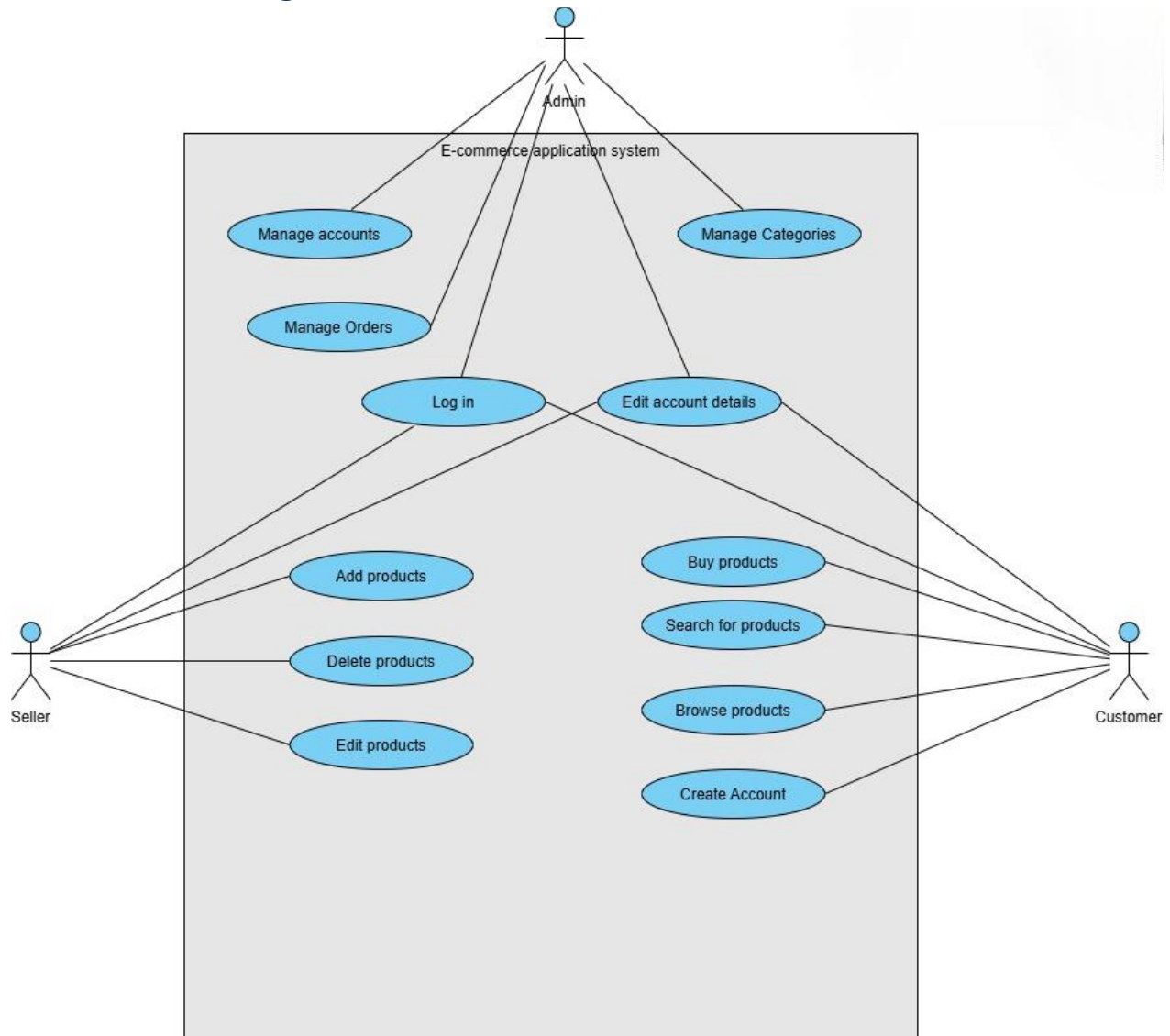
- User authentication and registration.
- Product listing, categorization, and management.
- Shopping cart and checkout system.
- Payment gateway integration.
- Order management and tracking system.

6.4 Non-functional Requirements

- **Reliability:** The system should be available all the time.
- **Usability:** System should be easy to use by having a simple graphical user interface.
- **Flexibility:** The system should be flexible enough to accommodate evolving data.
- **Accuracy:** Accurate results for clothing fitting on customers

7. System Analysis & Design

7.1 Use Case Diagram

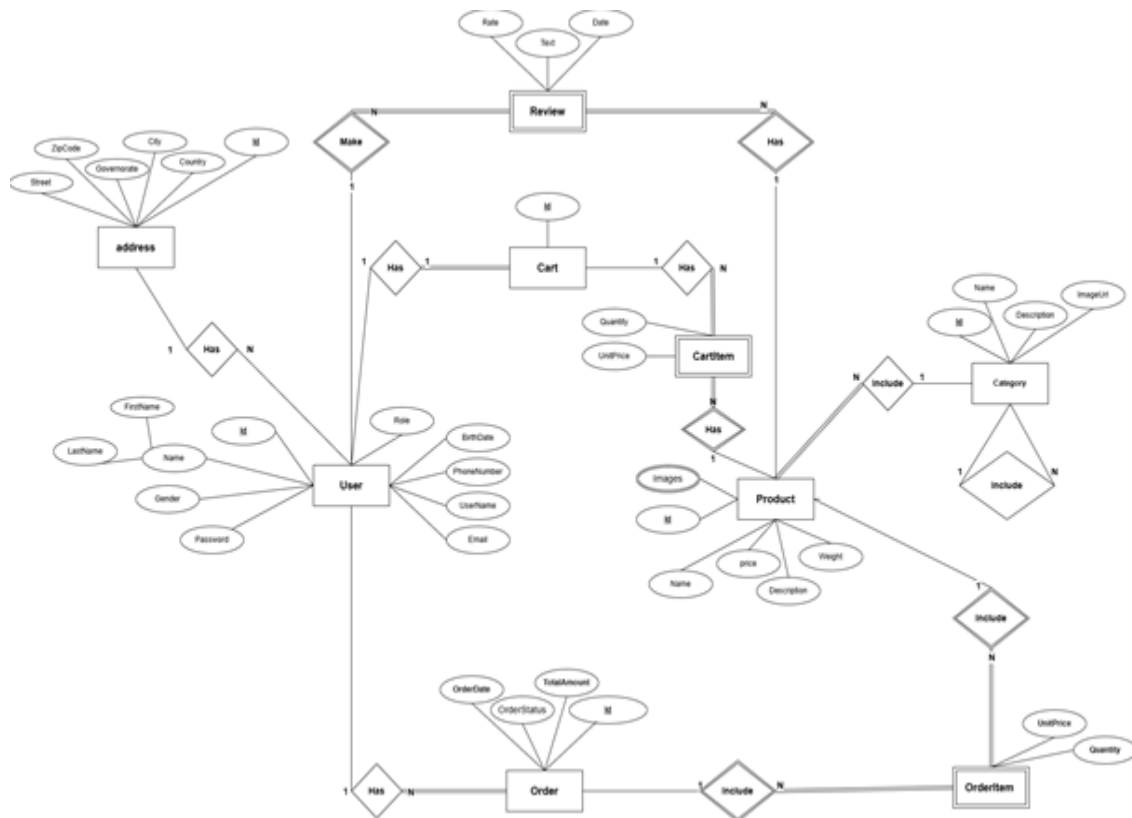


8. Software Architecture

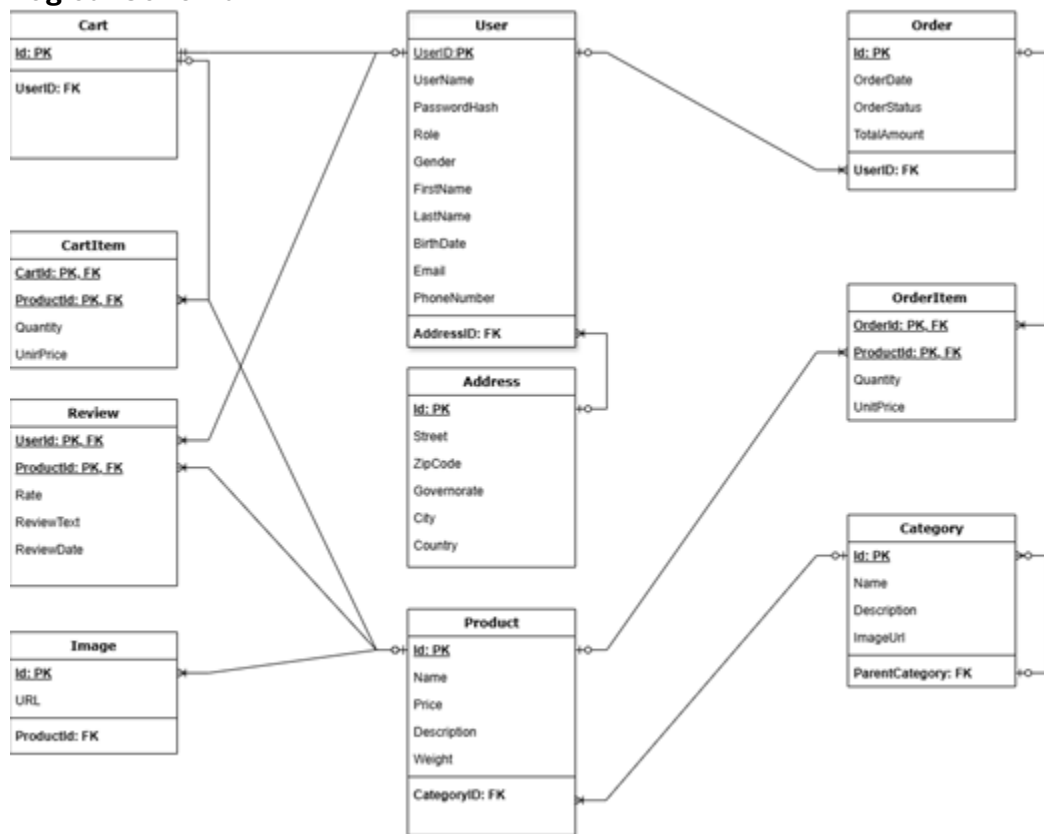
- Design system based on **n-tier Architecture**:
 - **Presentation Layer**
 - **Business Logic Layer**
 - **Data Access Layer**

8.1 Database Design

- ERD:



- **Logical Schema:**



9. Wireframes & Mockups

- Design search interface, product page, and shopping cart.

10. UI/UX Guidelines

- **Colors:** Neutral colors with blue accents for technology.
- **Typography:** Easy-to-read fonts.

11. Source Code

You can access the project repository on GitHub using the following link:

[LapStore GitHub Repository](#)

In this repository, you will find:

- **Frontend:** Develop using HTML, CSS, JavaScript.
- **Backend:** Develop using C# (.NET Core).
- **Database:** Design using SQL Server.

12. Security Measures

- Encrypt payment data using SSL.
- Validate inputs to prevent security attacks.

13. Unit & Integration Testing

- Test each component individually.
- Test integration between frontend and backend.

14. API Documentation

The LapStore platform provides a comprehensive RESTful API for all core functionalities. The API is organized into the following main categories:

- **Account Management:** User registration, login, profile management, and admin operations

Account		^
POST	/api/Account/Register	▼
POST	/api/Account/Login	▼
POST	/api/Account/Logout	▼
GET	/api/Account/Profile	▼
PUT	/api/Account/Profile	▼
POST	/api/Account/ChangePassword	▼
GET	/api/Account/Address	▼
POST	/api/Account/Address	▼
PUT	/api/Account/Address/{addressId}	▼
GET	/api/Account/admin/users	📄 ▼
GET	/api/Account/admin/users/{userId}	▼
DELETE	/api/Account/admin/users/{userId}	▼
PUT	/api/Account/admin/users/{userId}/role	▼
POST	/api/Account/admin/users/{userId}/disable	▼
POST	/api/Account/admin/users/{userId}/enable	▼
POST	/api/Account/admin/first-admin	▼

- **Shopping Cart:** Cart operations and item management

Cart		^
GET	/api/Cart	▼
DELETE	/api/Cart	▼
POST	/api/Cart/items	▼
PUT	/api/Cart/items/{productId}	▼
DELETE	/api/Cart/items/{productId}	▼

- **Product Catalog:** Product CRUD operations and search functionality

Product		^
POST	/api/Product	▼
GET	/api/Product	▼
PUT	/api/Product/{id}	▼
DELETE	/api/Product/{id}	▼
GET	/api/Product/{id}	▼

- **Order Processing:** Order creation, tracking, and status management

Order		^
GET	/api/Order	▼
POST	/api/Order	▼
GET	/api/Order/{orderId}	▼
GET	/api/Order/status/{status}	▼
PUT	/api/Order/{orderId}/status	▼

- **Categories:** Product categorization and filtering

Categories		^
GET	/api/categories	▼
POST	/api/categories	▼
GET	/api/categories/{id}	▼
PUT	/api/categories/{id}	▼
DELETE	/api/categories/{id}	▼
GET	/api/categories/exists	▼

- **Reviews:** Customer review and rating system

Review		^
GET	/api/Review/product/{productId}	▼
GET	/api/Review/user/{userId}	▼
POST	/api/Review/user/{userId}	▼
GET	/api/Review/user/{userId}/product/{productId}	▼
PUT	/api/Review/user/{userId}/product/{productId}	▼
DELETE	/api/Review/user/{userId}/product/{productId}	▼

